

PYTHON PROJECT

DATA ANALYSIS REPORT

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Dataset Name: Python_SalesData

OVERVIEW

This project demonstrated data analysis using Python (Pandas & Matplotlib) on raw sales data. The goal is to simulate the daily work of a data analyst.

- Clean the dataset (handle missing values, duplicates, formatting)
- Explore patterns and trends with descriptive statistics and aggregations.
- Answer key business questions.
- Provide insights and recommendations for business decisions.

The dataset contained sales records with information on products, quantities, revenue, payment methods, managers and locations.

DATA IMPORTATION

- Loaded the dataset into Pandas using: `pd.read_csv()`
- Previewed with `df.head()` and `df.info()` to understand columns and data types.

DATA CLEANING

- Checked for missing values.
- Removed duplicates.
- Removed spaces in text fields.
- Converted columns to their right data types.

DATA EXPLORATION

- Computed descriptive statistics.
- Used GROUP BY and aggregation for summaries.
- Created visualizations with Matplotlib to show patterns.

EXPLORATORY DATA ANALYSIS (EDA)

- Sales by product
- Sales by city
- Daily sales trends

BUSINESS QUERIES

1. Most Preferred Payment Method - Credit Card
2. i. Most selling product by quantity - Beverages
ii. Most selling product by revenue - Burgers
3. i. City with maximum revenue - Lisbon
ii. Manager with the maximum revenue - Joao Silva

4. Average revenue - 3029.5899
5. Average revenue in Nov & Dec - 3029.5899
6. i. Standard deviation of revenue - 2420.1183
ii. Standard deviation of quantity - 214.8886
7. i. Variance of revenue - 5856972..9637
ii. Variance of quantity - 46177.1530
8. Trend analysis: The revenue is increasing over time
9. Average quantity sold and average revenue for each product.
10. Total number of orders - 254

VISUALIZATIONS

These visuals made the findings clearer and easier to interpret

- Revenue by product
- Quantity sold by product
- Revenue trend over time
- Most preferred payment method
- Revenue by city
- Revenue by manager

INSIGHTS

- Customers prefer Credit card payments
- Revenue is concentrated in Lisbon under Manager Joao Silva.
- Revenue shows a seasonal spike in November and December.
- Beverages sell the most by volume but Burgers brings in more revenue.
- A small number of managers drive the majority of revenue.

RECOMMENDATIONS

- Customer Preferences
Ensure smooth seamless processing and offer loyalty rewards.
- Geographic Sales Trends
Replicate their successful sales strategy in the weaker cities.
- Revenue and Seasonality
Plan marketing campaigns earlier to capture demand.
Increase inventory, staffing and promotions.
- Product Performance
Ensure steady inventory for product Beverages to avoid stock outs.
Consider bundling both products (Beverages & Burgers) together to maximize sales.